

Radiation and Mass Transfer Effects on MHD Oscillatory Flow of a Chemically Reacting Fluid in a Channel Filled with Porous Medium

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Abstract

This paper investigates the effect of thermal radiation on an unsteady heat and mass transfer flow of a viscous incompressible electrically conducting fluid through a channel filled with saturated porous medium, taking homogeneous chemical reaction of first order into account. The fluid is assumed to be optically thin and the Cogley diffusion approximation is used to analyze the radiative heat flux in the energy equation. This model finds applications in the design of high temperature chemical process systems.

Keywords: Oscillatory flow, porous medium, heat and mass transfer, magnetic field, radiation, chemical reaction.

Introduction

The study of flow problems, which involve the interaction of several phenomena, has a wide range of applications in the field of science and Technology. One such study is related to electrically conducting viscous fluids within channels or ducts which are important from a practical point of view in magnetohydrodynamic accelerators, hydromagnetic lubrication, re-entry astronautics, magnetic storage disks, metallurgical systems, etc. The effect of transverse magnetic field on the flow of electrically-conducting fluid in a two-dimensional channel has been treated theoretically by Hartmann [1]. He obtained an exact solution of the MHD equations for a viscous incompressible fluid. Hydromagnetic reactive flows are often